

SMART INDIA HACKATHON 2026



AAMS-X

Adaptive Autonomous Mission Spectrum eXplorer

An explainable spectrum research workbench

Problem Statement ID	SIH26055
Problem Statement Title	Smart Scan strategy for Electronic Warfare
Theme	Awaiting portal confirmation
PS Category	Software
Team ID	Not supplied
Team Name (Registered on portal)	AlphaX



Proposed solution

A local workbench to test and explain adaptive scan decisions when the receiver can observe only part of the spectrum.

The problem

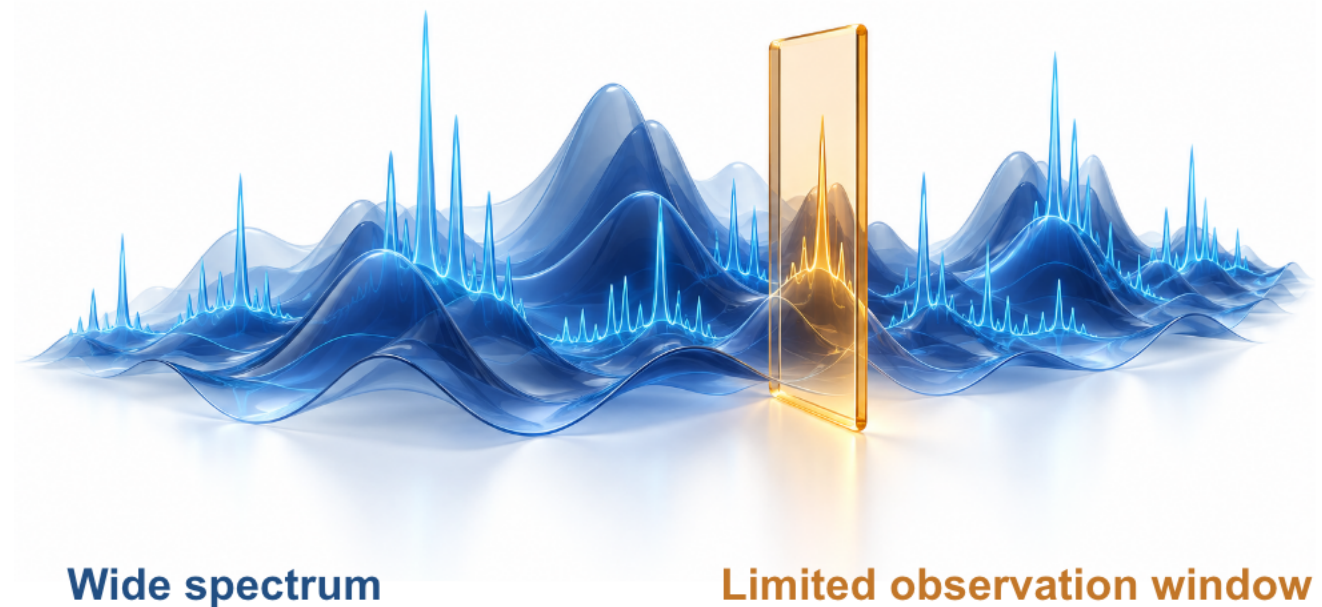
A narrow observation window leaves most of the spectrum unknown at each instant.

How AAMS-X addresses it

Compare fixed and adaptive policies in the same simulated world, then inspect each recorded decision.

Innovation and uniqueness

One auditable workflow combines live comparison, explanations, provenance and exact recorded replay.



Wide spectrum

Limited observation window

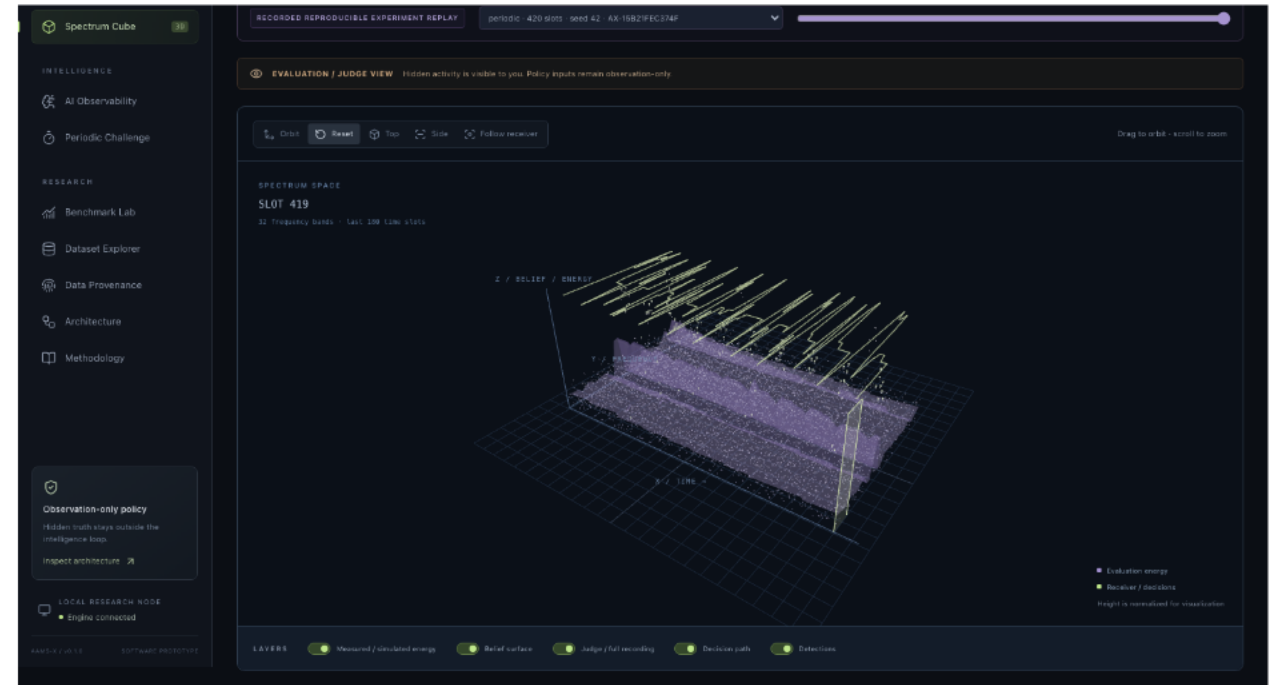
Concept illustration, not measured data

Methodology and implementation

- 01 Reproducible inputs**
Seeded simulations or a public RF recording.
- 02 Observation boundary**
Only the selected window reaches the policy.
- 03 Inspectable model state**
Recorded beliefs and explanations stay visible.
- 04 Independent evaluation**
Compare policies and save verified traces.

Technologies

Python / FastAPI / NumPy / SciPy
React / TypeScript / Three.js / SQLite



Working prototype: recorded 3D exploration

Evaluation view shown. Hidden truth stays outside policy inputs.



FEASIBILITY AND VIABILITY



A working software prototype with reproducible evidence

32

backend tests passed

30

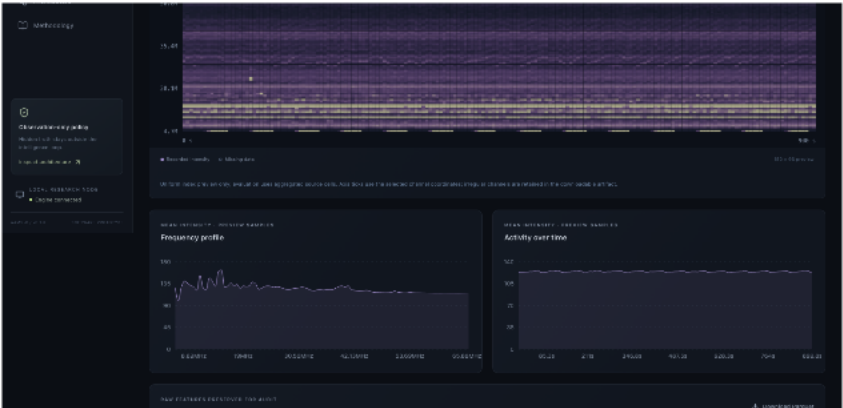
paired simulation worlds

180

policy evaluations

Challenges and validation strategy

Current limitation	How AlphaX will address it
Official TSRD absent	Obtain authorized data, then verify the import.
Hardware unvalidated	Plan independent calibration and lab review.
Scenario-dependent results	Expand seeds and independent test datasets.



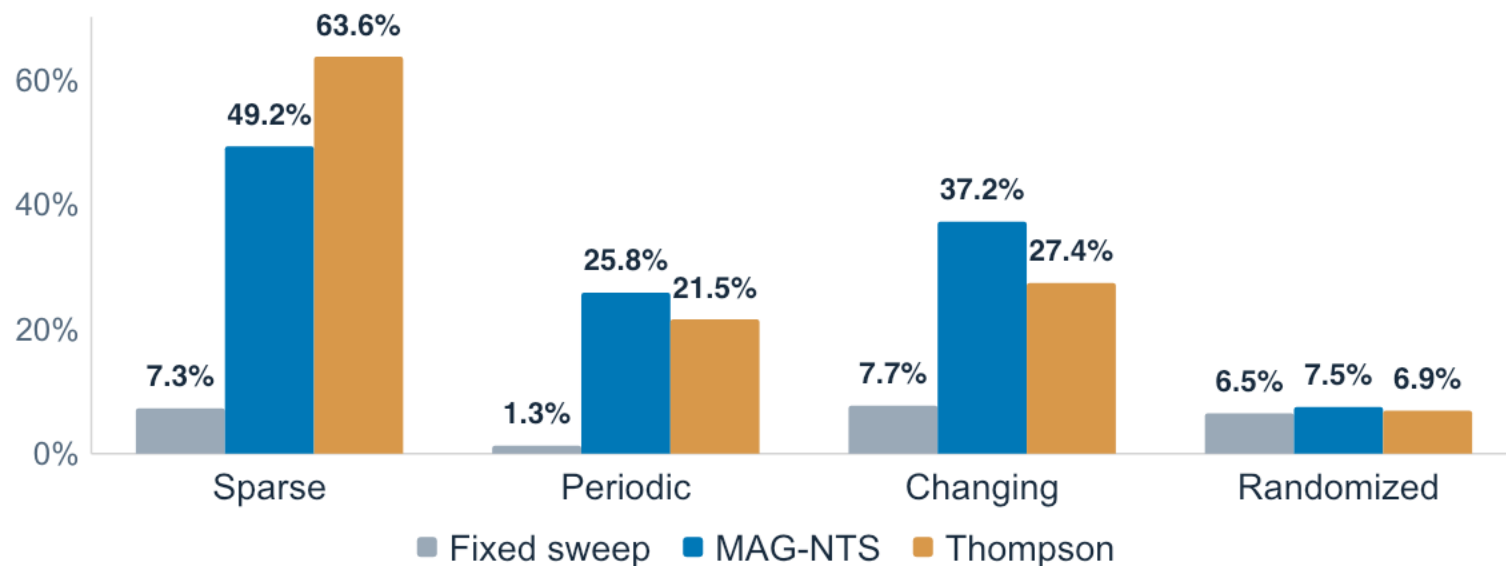
Measured data already included

705,600 standardized solar-radio samples.
No labelled EW emitter ground truth.

Viability: local operation after setup, open-source libraries and no paid API dependency.

Simulation evidence shows both gains and limits

Global recall: detected active cells / all active band-time cells



Selected scenarios. Means over seeds 42–44, 240 slots. Full suite: 10 scenarios and 6 policies.

Thompson leads in the sparse case. Three seeds do not establish universal superiority or field performance.

For research reviewers

Replay the same evidence and inspect assumptions behind each comparison.

For engineering teams

Identify software and data gaps before proposing hardware trials.

Expected benefit

More transparent evaluation.
Cost and field benefits remain unmeasured.

Research foundations and auditable project evidence

01 Russo et al. (2018)

A Tutorial on Thompson Sampling

arxiv.org/abs/1707.02038

Foundations for learning under uncertainty.

02 Gunn et al. (2026)

The Turing Synthetic Radar Dataset

arxiv.org/abs/2602.03856

Importer implemented. Authorized data still required.

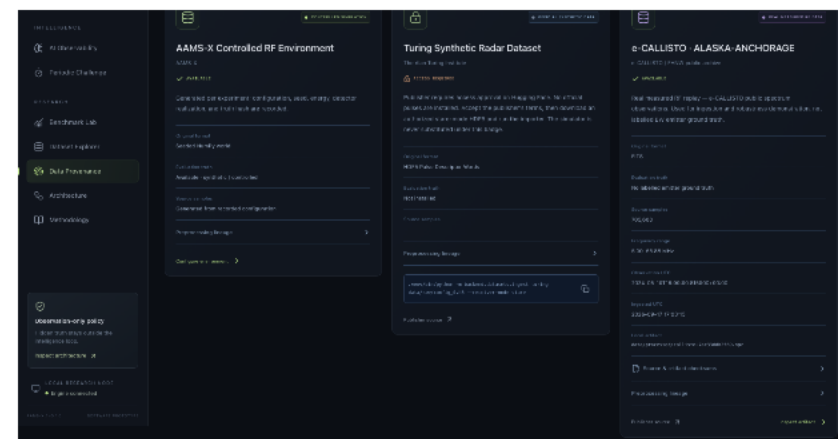
03 e-CALLISTO / FHNW

Public solar-radio observations

www.e-callisto.org/

Measured recording included with source lineage.

AlphaX evidence package



BENCH-723C785465

Checksummed traces, exact replay,
metric definitions and source manifests.

Reproduce locally: `./start_demo.sh`

Project evidence: README.md, docs/DATA.md, docs/METHODOLOGY.md and the bundled benchmark report.